

Urban visual intelligence: Uncovering hidden city profiles with street view images

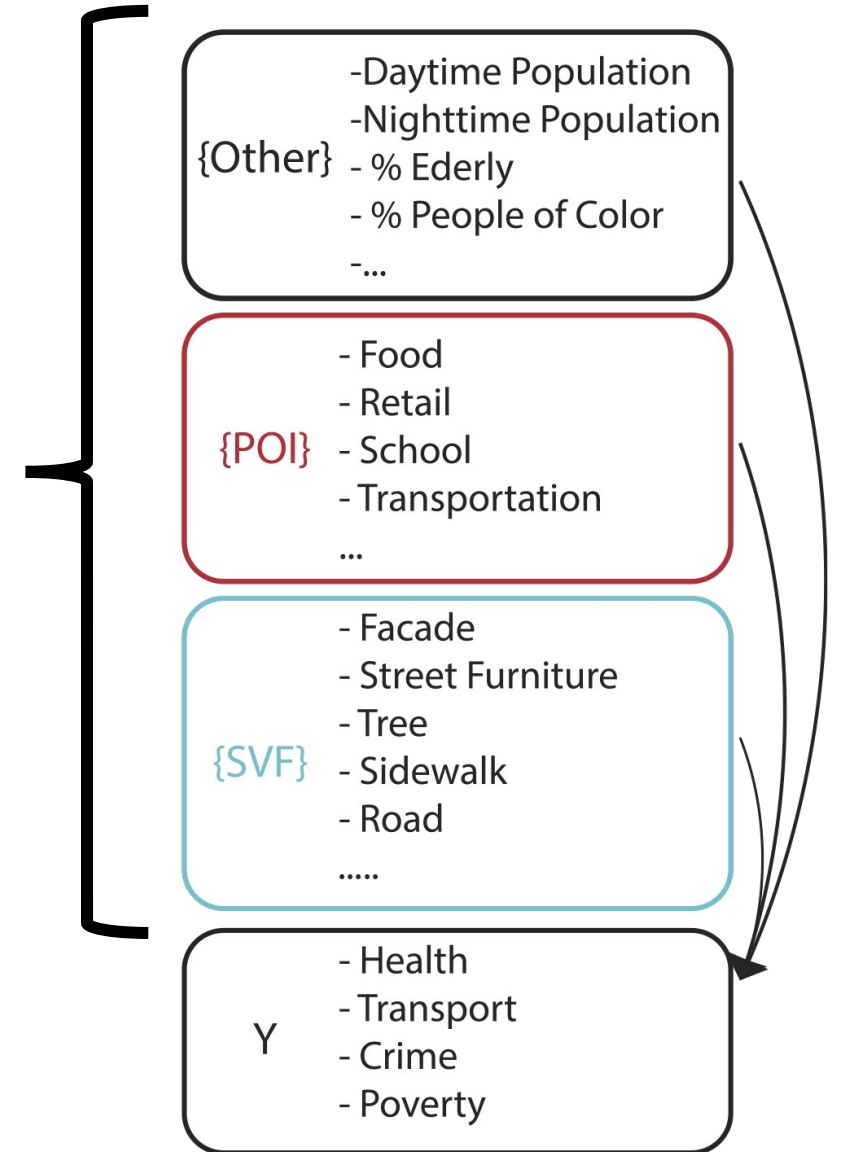
Zhuangyuan Fan , Fan Zhang, Becky P. Y. Loo,
Carlo Ratti

Proceedings of the National Academy of Sciences of the United States of America (2023)

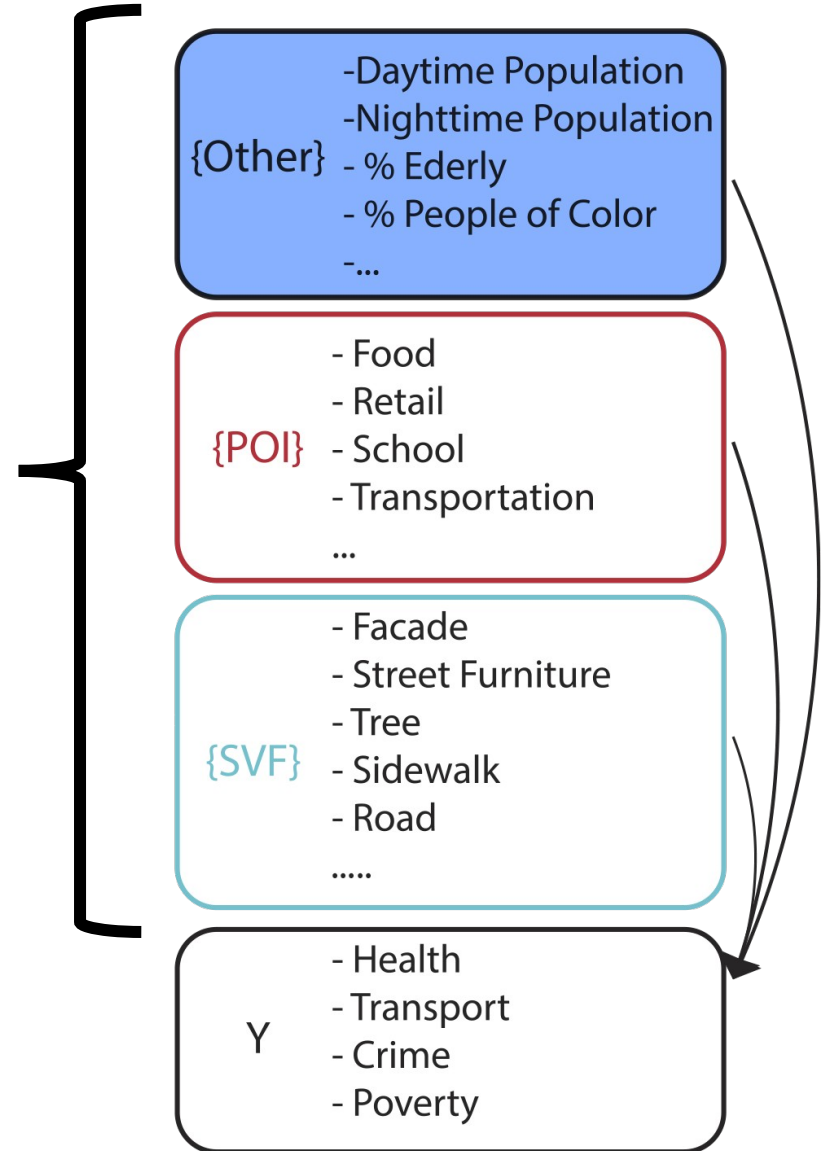
Presented by Haoanqin Gao

- Y
- Health
 - Transport
 - Crime
 - Poverty

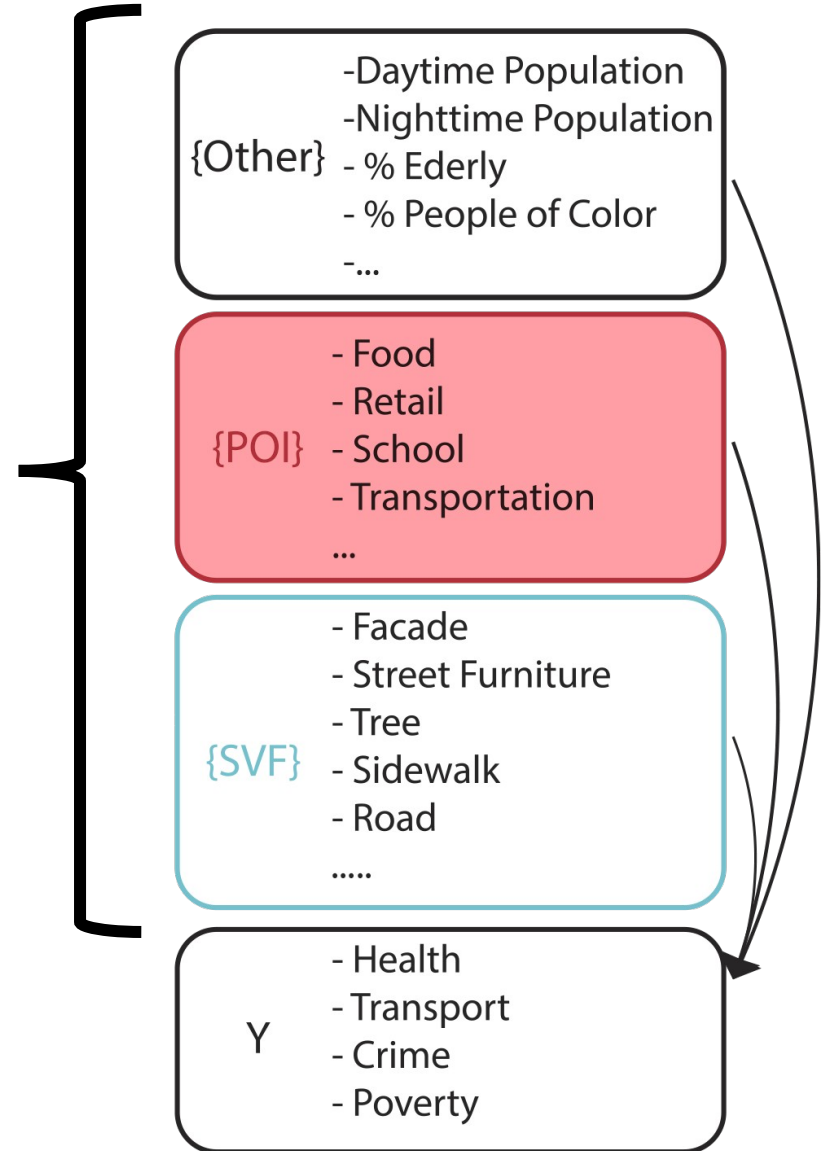
3 feature types
are compared



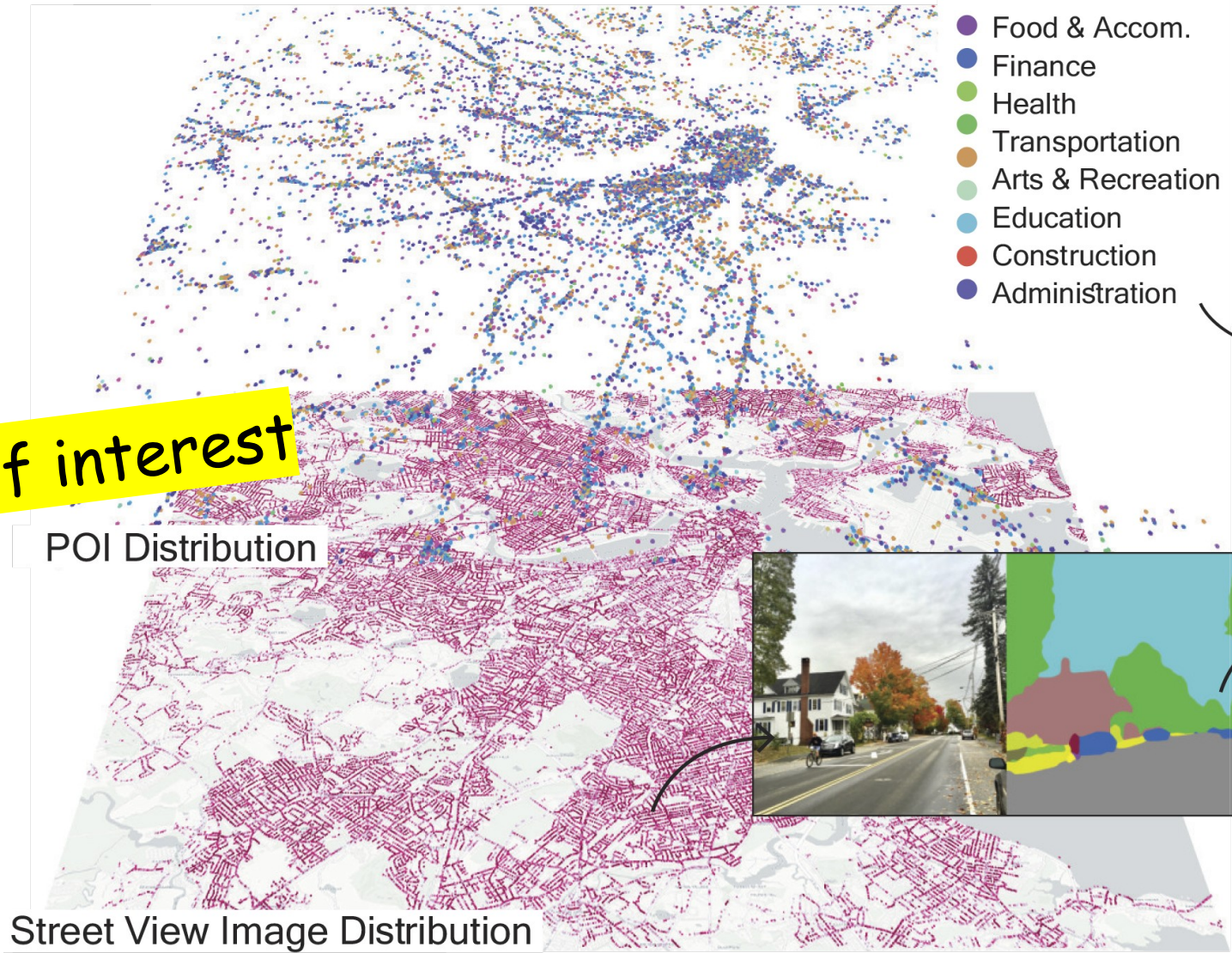
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point of interest

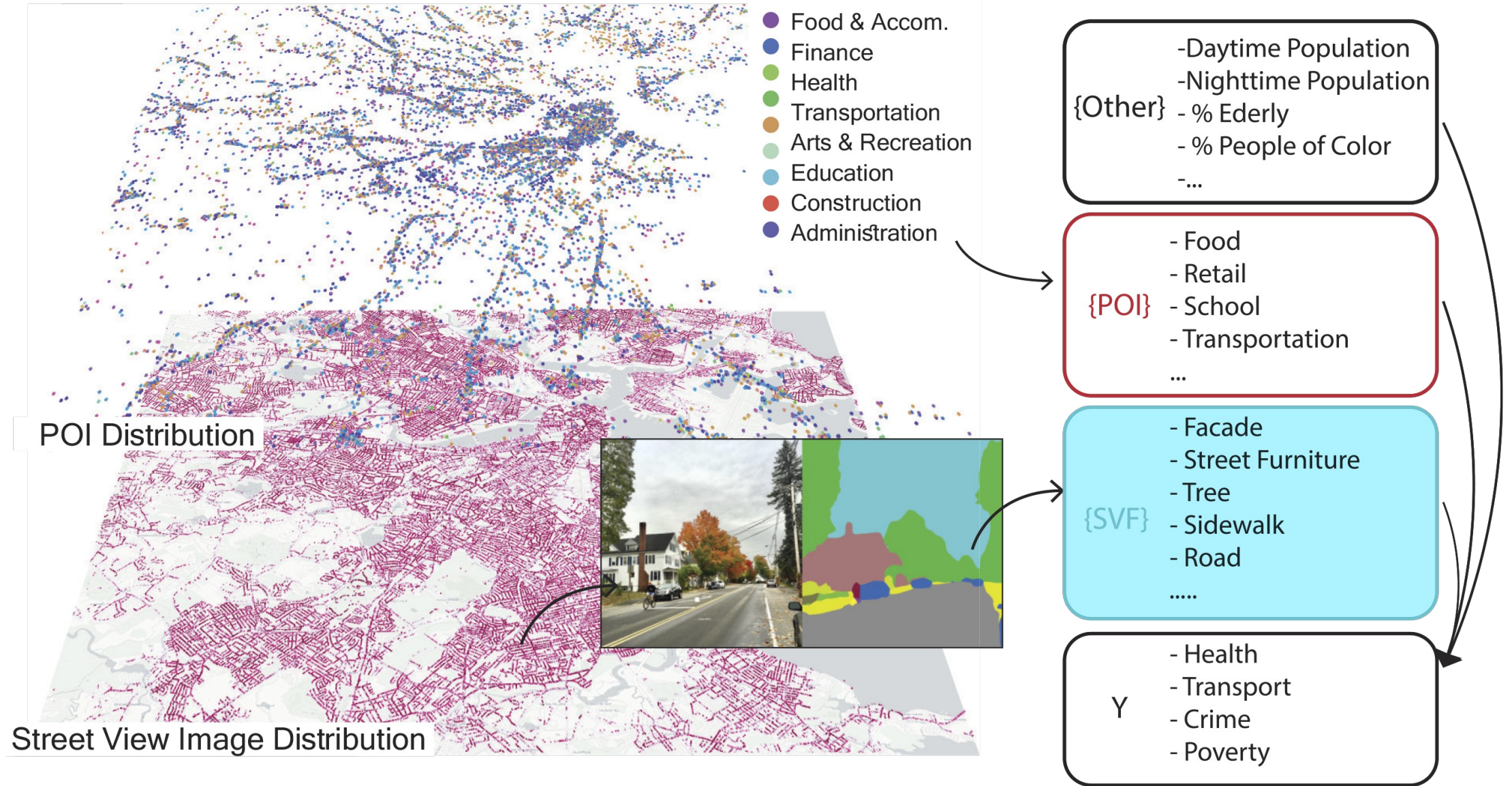


- {Other}
- Daytime Population
 - Nighttime Population
 - % Elderly
 - % People of Color
 - ...

- {POI}
- Food
 - Retail
 - School
 - Transportation
 - ...

- {SVF}
- Facade
 - Street Furniture
 - Tree
 - Sidewalk
 - Road
 -

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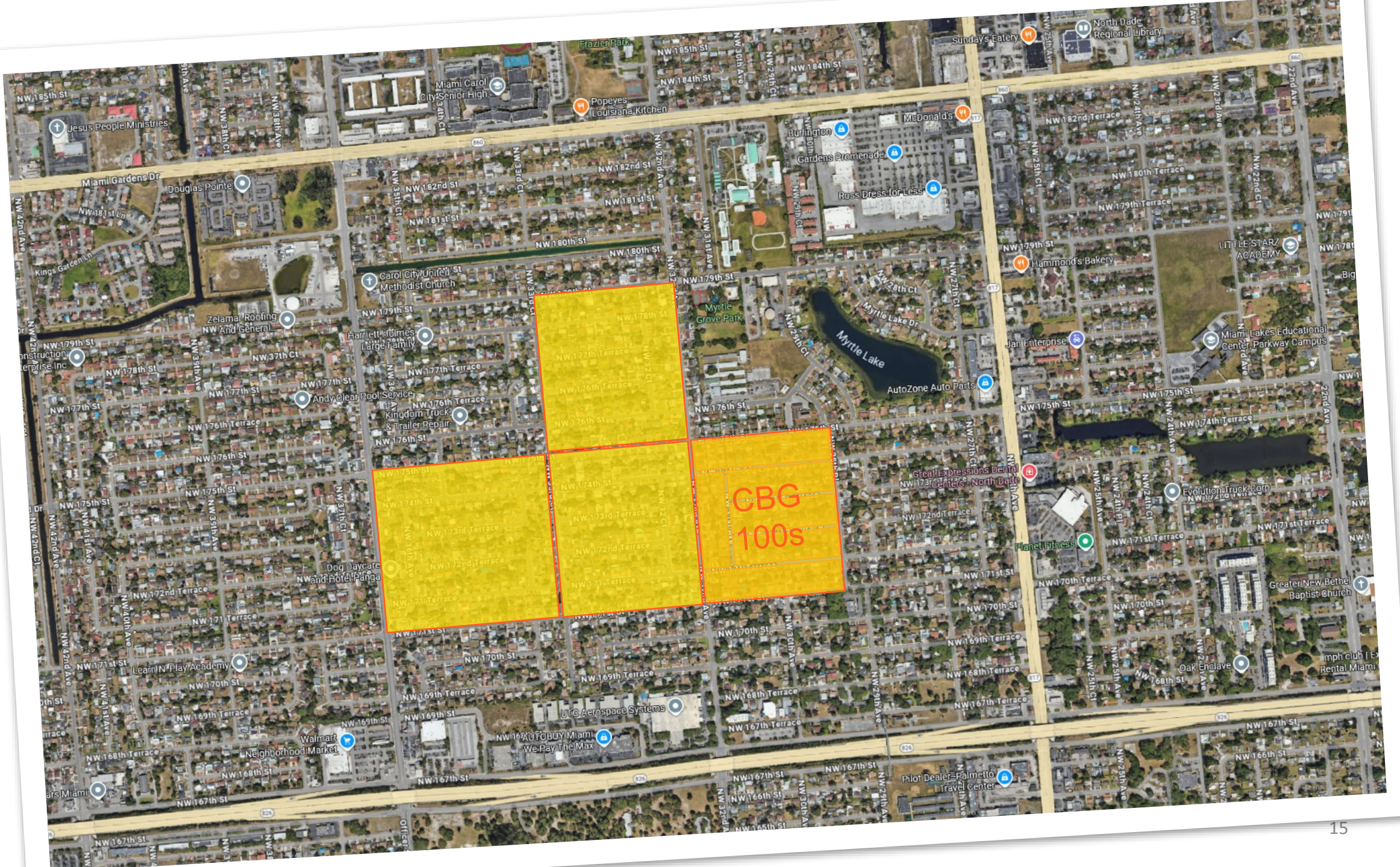
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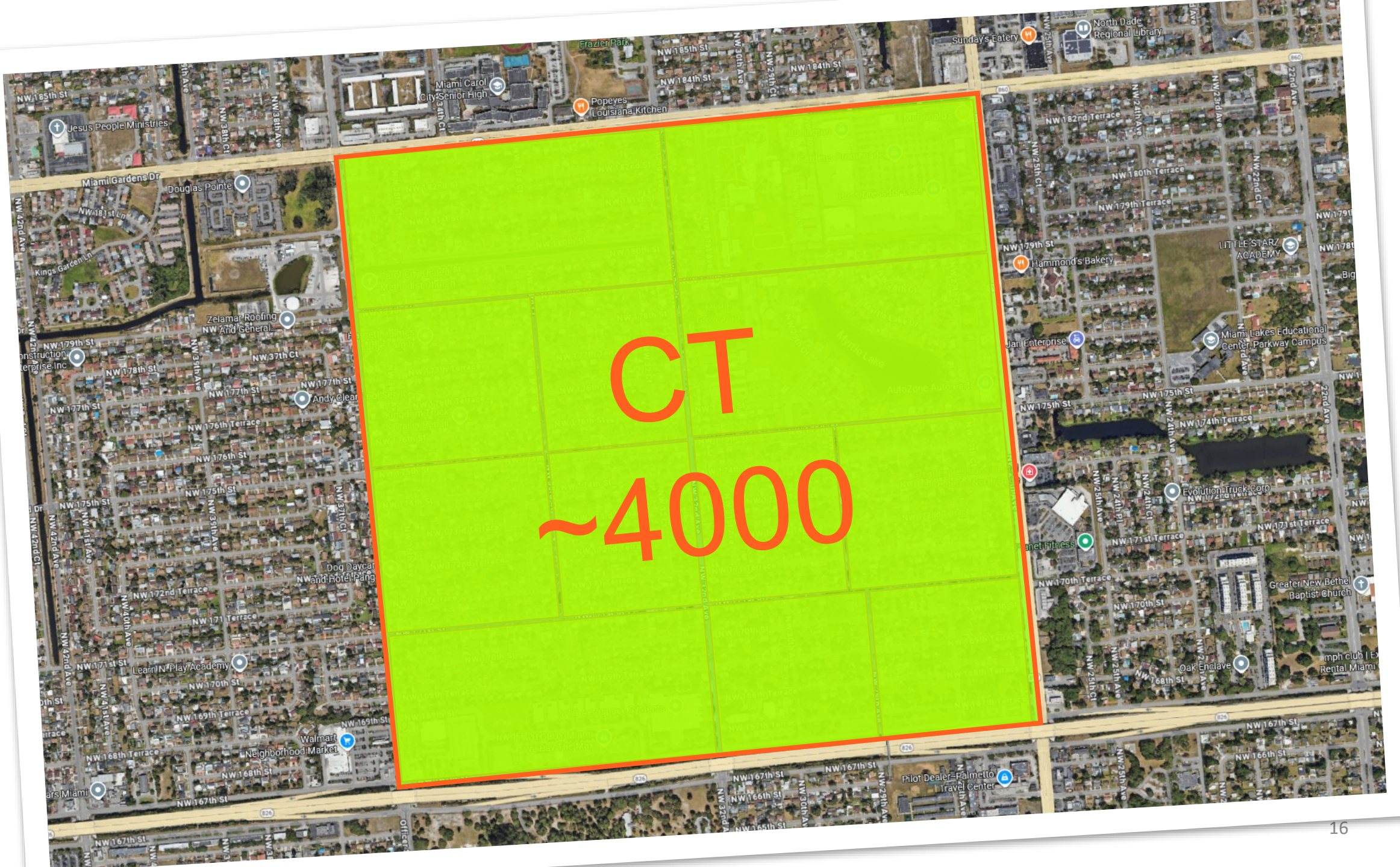
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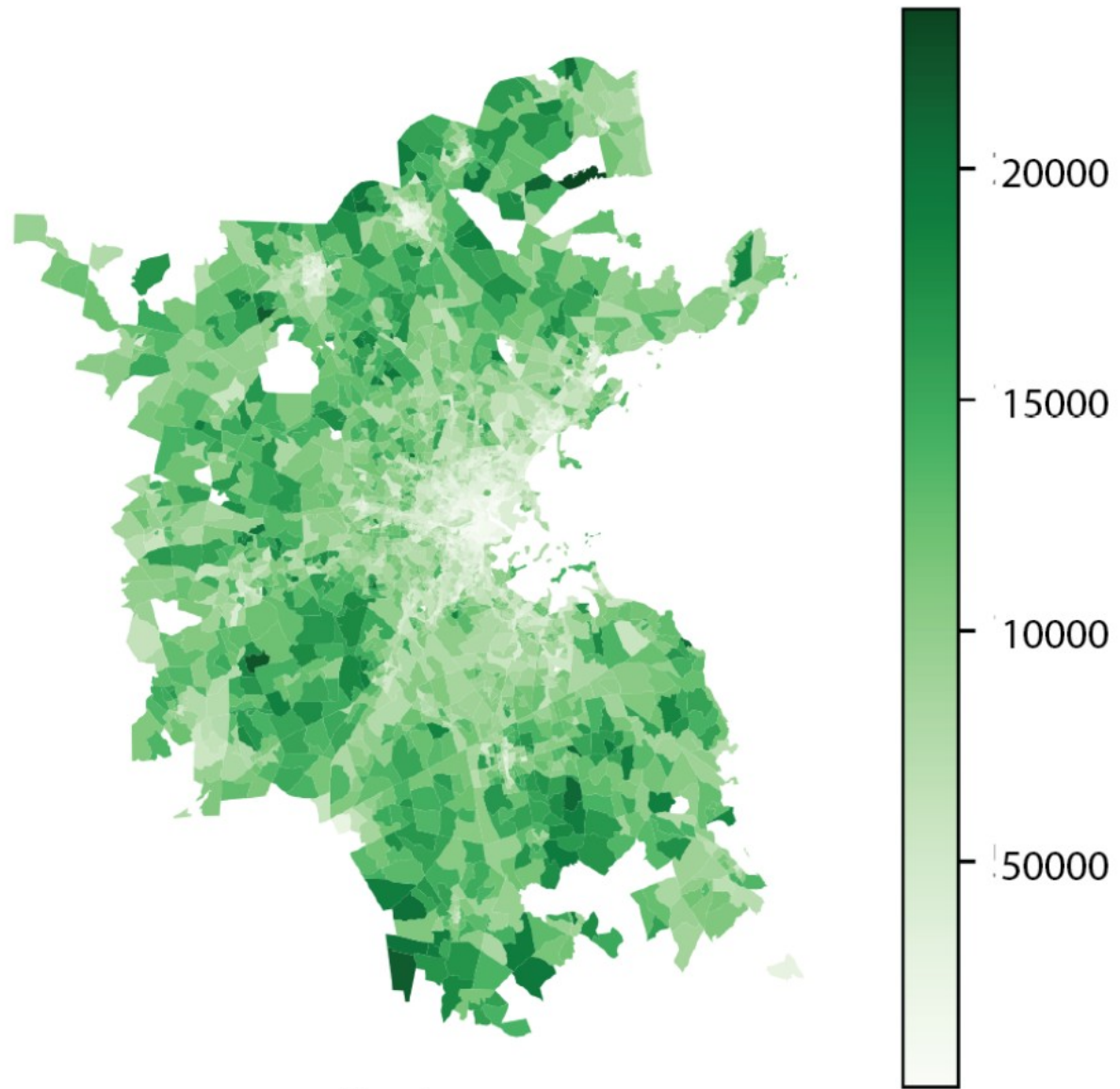






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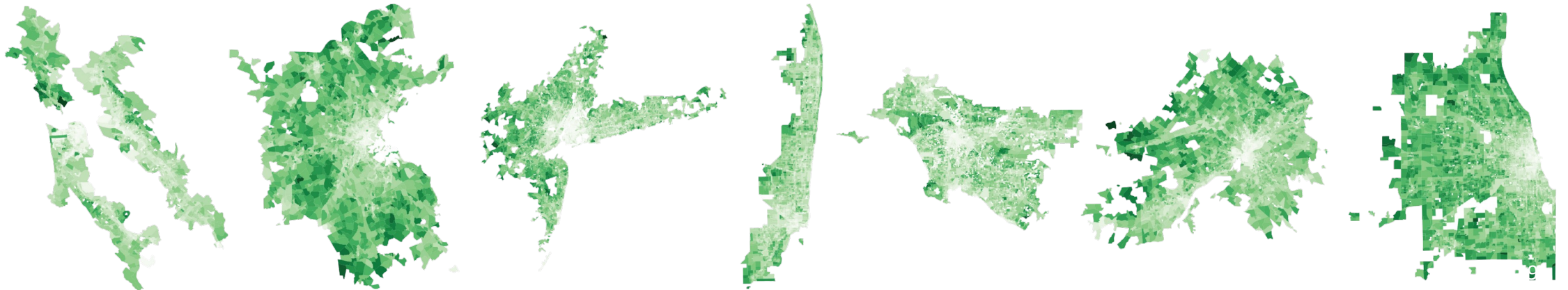


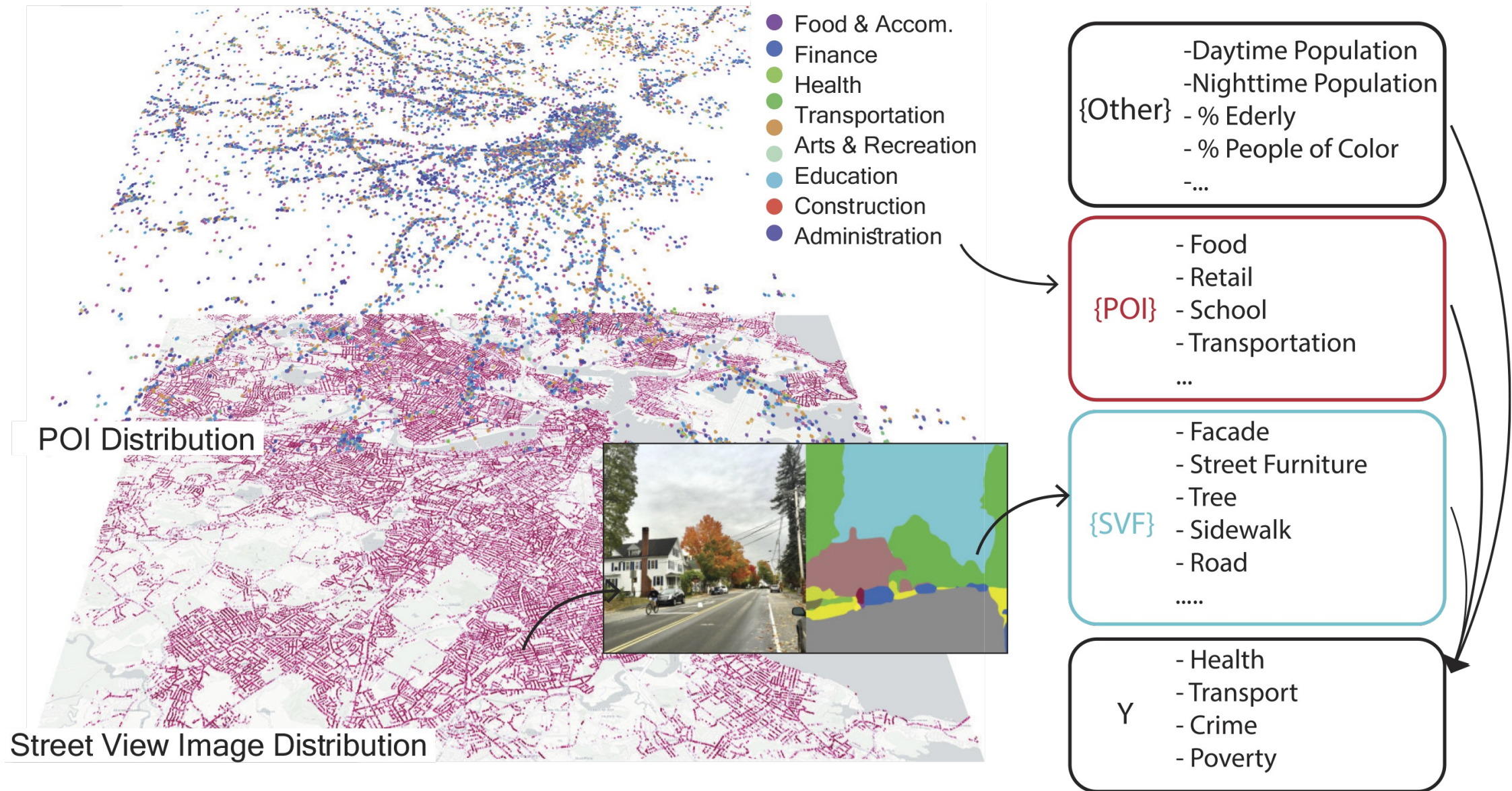
Boston

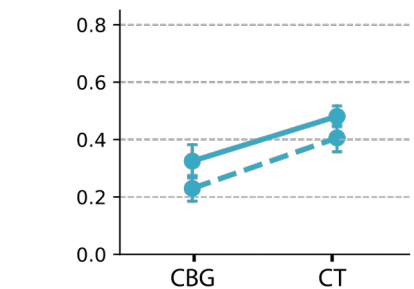
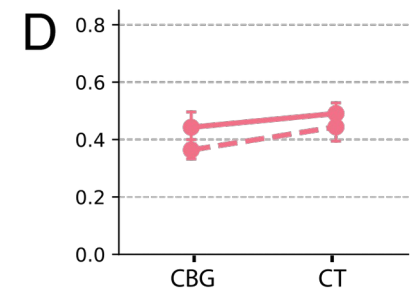
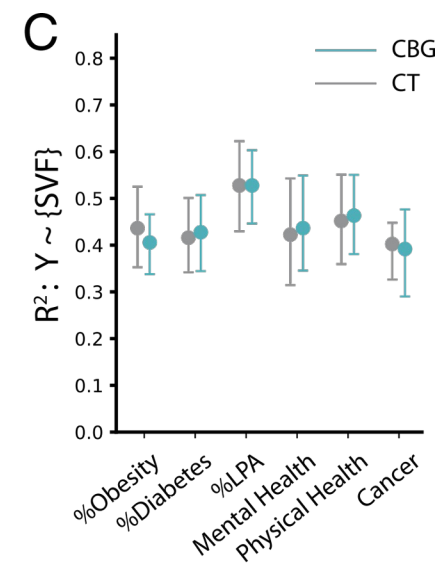
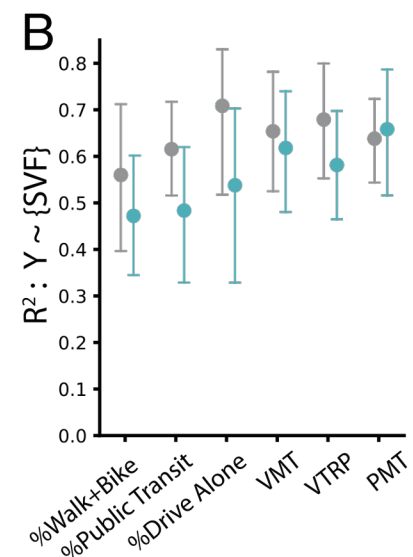
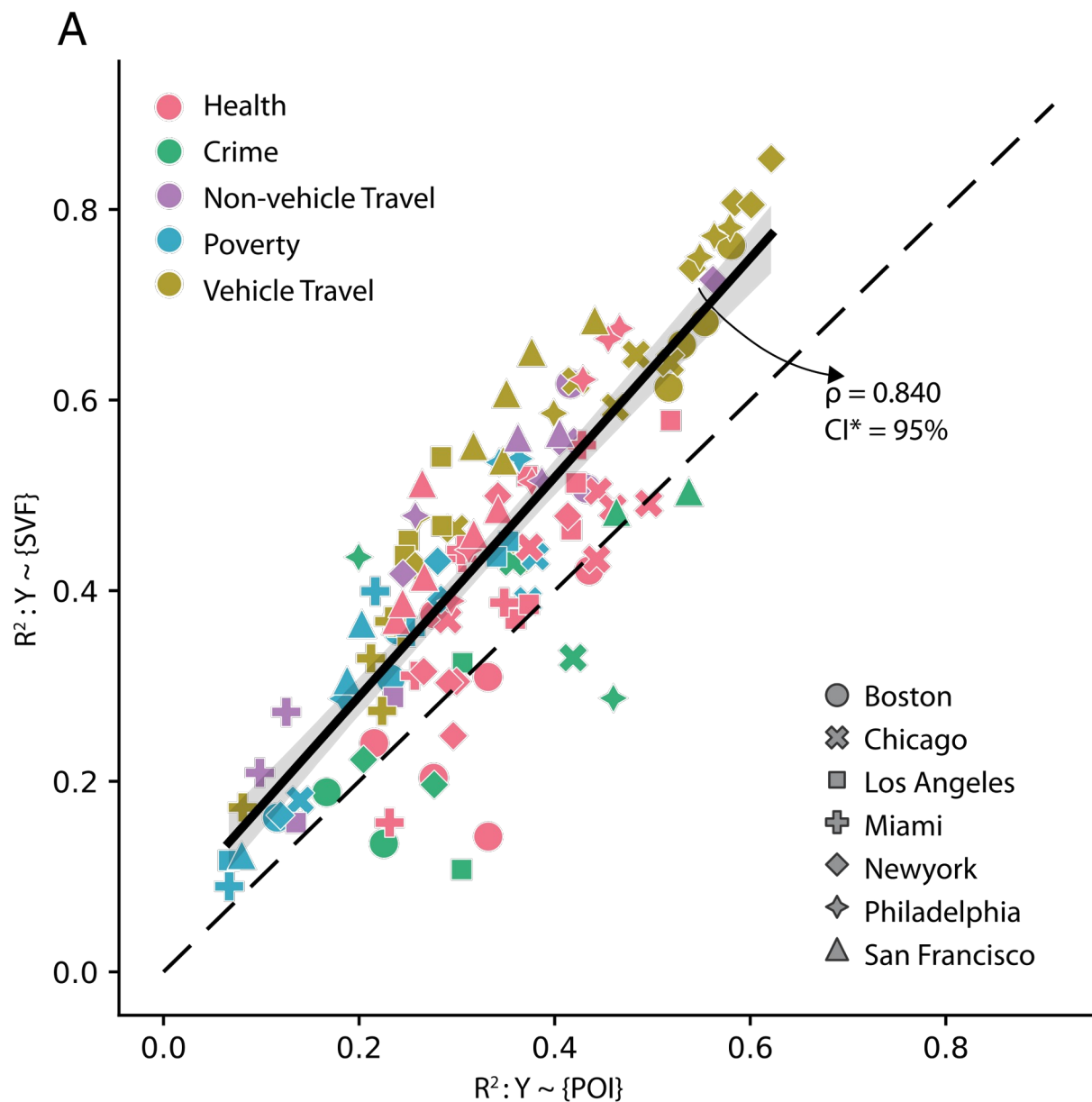
distribution of grass

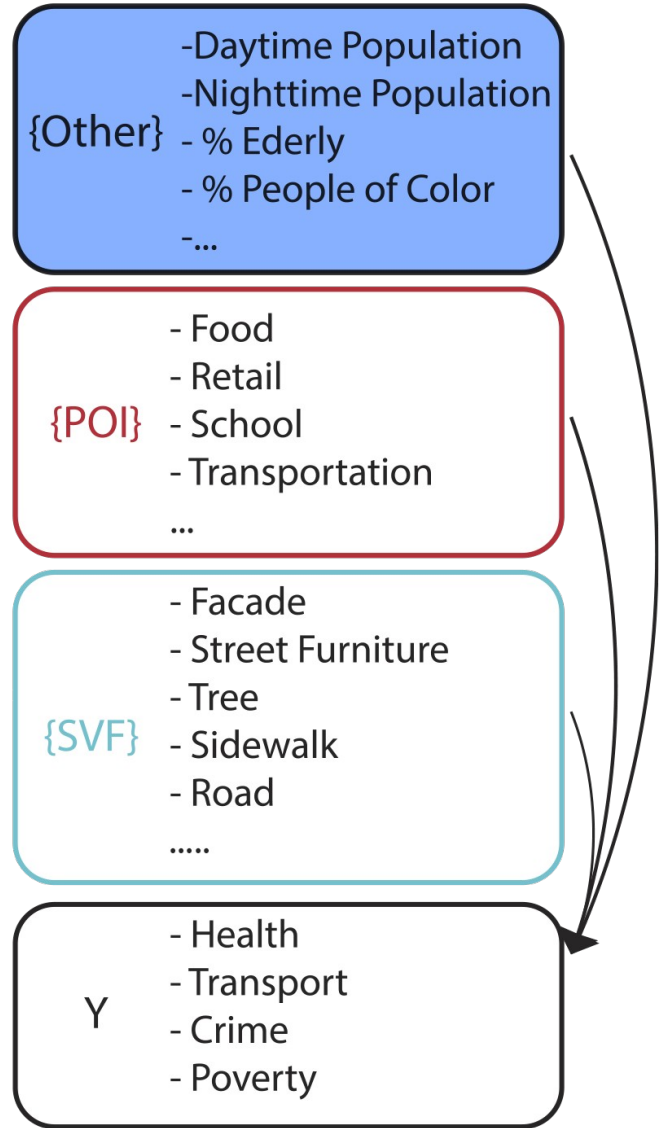
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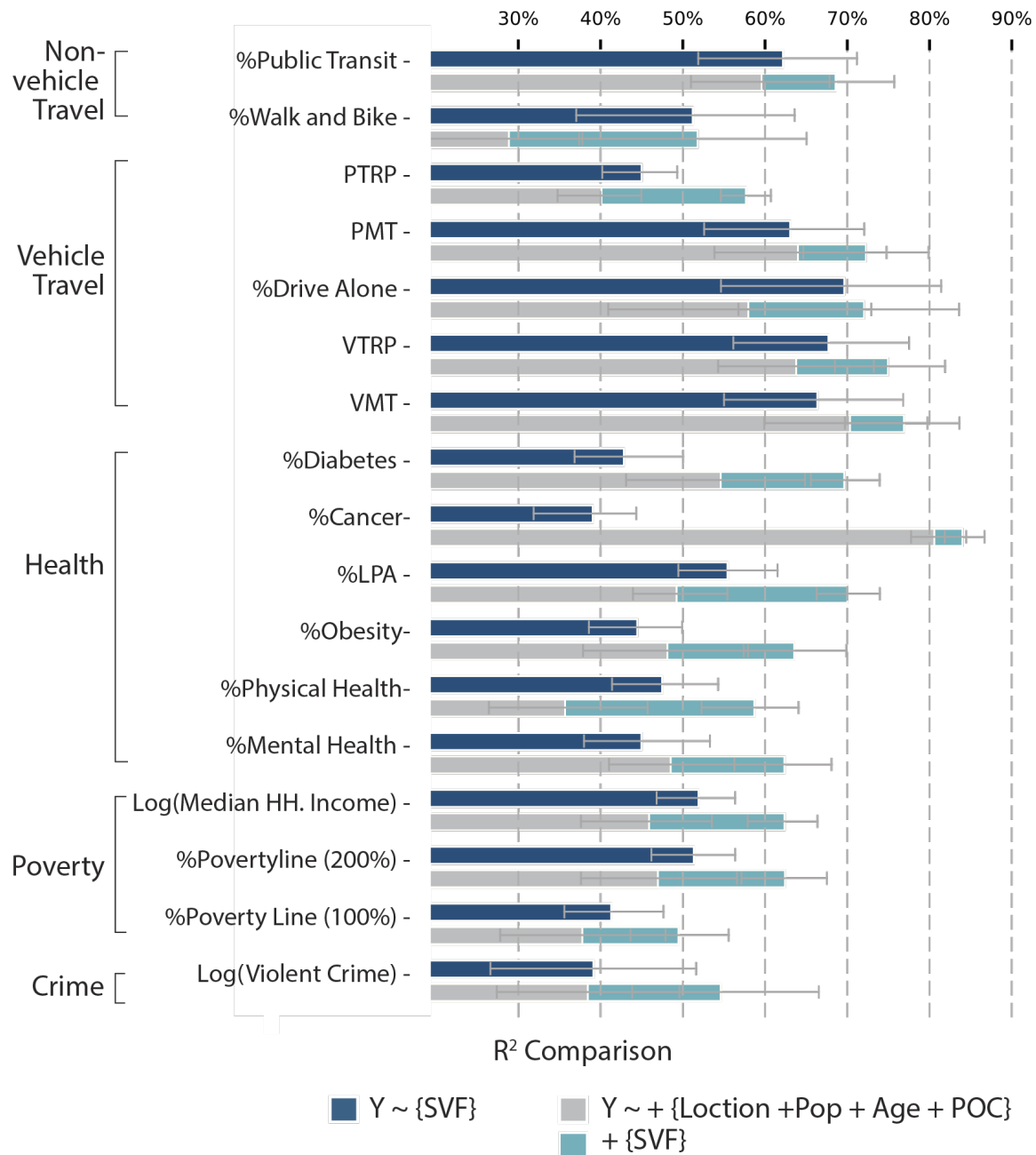
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Street View Features...

- estimate travel behaviors particularly well
- are a poor estimator of certain health outcomes
- outperform POI and dynamic populations
- are limited by the coverage of Google Street View
- complement other available statistics to deliver better estimation

Discussion questions

- Given the prediction power of street view images, should a single company like Google be trusted with this much data?
- Can SVF really help local planners design better cities? Is it not simply imitating what humans are naturally capable of?